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Timing and Tempo of Puberty and Neurodevelopment Following Adversity: A Registered Report

--Manuscript Draft--

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Abstract:	Theoretical models suggest that stress exposure may influence the pace of development, though empirical findings are mixed. Some research links adversity to delays in brain and pubertal development, while other studies find accelerated maturation. We will use seven waves of data from the Adolescent Brain Cognitive Development (ABCD) Study (n = 11,868) to examine associations between adversity and the timing and tempo of pubertal and brain development, as well as working memory. We hypothesize that higher adversity will predict more advanced pubertal and brain maturation at baseline, which will in turn be associated with higher working memory. Pubertal timing and tempo will be modeled using logistic growth curves, while subcortical and cortical gray matter volume (GMV) intercepts and slopes will be estimated via latent growth curve models. Structural equation modeling will test how adversity relates to pubertal and GMV trajectories, and their associations with working memory. Separate models will be estimated for males and females. These analyses aim to advance understanding of how adversity shapes variability in developmental trajectories, with implications for supporting and optimizing healthy adolescent development.
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November 18, 2025

Dear Dr. Hanson,

Thank you for the opportunity to submit a second revision of our Stage 1 registered report entitled, *Timing and Tempo of Puberty and Neurodevelopment Following Adversity: A Registered Report.* We have responded to the remaining minor reviewer comments.

We agree to openly share all materials from the project, including analytic code. The data are available to users with an active data use agreement. Following Stage 1 in principle acceptance, we agree to register the approved protocol on the Open Science Framework. We confirm that if we later withdraw the manuscript that the journal may publish a short summary of the pre-registered study under "Withdrawn Registrations".

We look forward to your response. Thank you for your time and consideration.

Sincerely,

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Response to Reviewers

Comments from the Editors and Reviewers: Thank you for resubmitting this manuscript. Reviewers continued to see clear value in the work and it is obvious that the revised submission has improved significantly from the initial submission. One reviewer raised some small concerns and they are noted below. My hope is that your team can thoughtfully attend to these and I can accept the work without needing to send this manuscript back out for further review.

We thank the editor and reviewers for their careful consideration of our manuscript. We have responded to the final comments point-by-point below. The specific reviewer comments are restated, and our responses are bolded. All associated changes have also been made in the manuscript; those edits have been made in track changes. We look forward to working on the Stage 2 registered report.

Reviewer #1:
The authors were responsive to my critiques.

Reviewer #2:
I thank the authors for their comprehensive responses to my comments and related revisions.

1. My remaining concern is in relation to my earlier point 4. From the updated text, it is still not clear to me why the authors would expect early puberty to relate to better cognition, given they report evidence for both enhanced and reduced cognition, potentially dependent on the time of assessment.

Youth in ABCD are approximately 15-16-years-old at the last point of data collection. Thus, it is likely that many, *but not all,* youth will have completed puberty. We expect that early puberty will be associated with an initial "cognitive boost" relative to on-time and later-maturation. This pattern is consistent with past research, suggesting that when some youth have reached more advanced stages of pubertal maturation but others have not, early developers temporarily outperform their peers who are in earlier stages of development (Chaku & Hoyt, 2019; Torvik et al., 2021). However, evidence from studies with longer time frames suggest that on-time or later developers may "catch-up" or exceed the cognitive performance of early developers (Goering et al., 2023; Juraska & Willing, 2017). We cannot explicitly test whether this is an age-dependent effect or an enduring cognitive difference because not all youth have completed puberty. We will discuss this possibility in the limitations and future directions section of the manuscript.

Introduction, p 5-7: "Earlier timing has been associated with both accelerated and delayed cognitive development as well. Some studies report enhanced cognitive control and working memory skills in earlier developers* (Chaku & Hoyt, 2019; Torvik et al., 2021) *and others report worse cognitive outcomes in earlier developers, especially later in adolescence as their peers "catch-up" developmentally* (Goering et al., 2023; Juraska & Willing, 2017)*. It is yet unclear whether earlier developers experience an initial cognitive

boost that dissipates over time, whether effects are domain-general or domain-specific (see Luna et al., 2015; Del Giudice & Crespi, 2018) or whether slower, more protracted maturation ultimately supports more optimal cognitive development."

2. Also, please define what your "families of related tests" will be, as this is open to interpretation.

"Families of related tests" refers to conceptually and statistically related sets of hypotheses that address the same analytic question and are estimated within the same model. See p 20 for our updated text:

"To control for multiple comparisons, p-values from planned tests will be adjusted using the Benjamini–Hochberg False Discovery Rate (FDR) procedure (Benjamini & Hochberg, 1995). The FDR procedure will be applied separately within each family of related tests (RQ1-RQ3: path analyses, R4: parallel mediation analysis, RQ5: Wald tests) separately for girls and boys using a false discovery rate of 0.05."

Data Statement

Data from the Adolescent Brain Cognitive Development (ABCD) Study are available to investigators with an active data use agreement. Data access and information about the 6.0 release can be found at: https://doi.org/10.82525/jy7n-g441

Marked up Manuscript Click here to view linked References

Timing and Tempo of Puberty and Neurodevelopment Following Adversity: A Registered Report

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Abstract

Theoretical models suggest that stress exposure may influence the pace of development, though empirical findings are mixed. Some research links adversity to delays in brain and pubertal development, while other studies find accelerated maturation. We will use seven waves of data from the Adolescent Brain Cognitive Development (ABCD) Study (n = 11,868) to examine associations between adversity and the timing and tempo of pubertal and brain development, as well as working memory. We hypothesize that higher adversity will predict more advanced pubertal and brain maturation at baseline, which will in turn be associated with higher working memory. Pubertal timing and tempo will be modeled using logistic growth curves, while subcortical and cortical gray matter volume (GMV) intercepts and slopes will be estimated via latent growth curve models. Structural equation modeling will test how adversity relates to pubertal and GMV trajectories, and their associations with working memory. Separate models will be estimated for males and females. These analyses aim to advance understanding of how adversity shapes variability in developmental trajectories, with implications for supporting and optimizing healthy adolescent development.

Keywords: adolescence; adversity; working memory; puberty; longitudinal; neurodevelopment

Timing and Tempo of Puberty and Neurodevelopment Following Adversity

Exposure to adverse experiences and environments early in life can accumulate and cascade in interactive ways across developmental domains (Masten & Cicchetti, 2010). During adolescence, these effects may become especially pronounced, given ongoing physical and neurobiological maturation. Theoretical models suggest that adversity exposure can alter the timing and pace of developmental processes (Belsky & Shalev, 2016; Callaghan & Tottenham, 2016), but empirical findings have been mixed. Some studies suggest that stress and adversity exposure is associated with patterns of acceleration in pubertal and brain development (e.g., Belsky et al., 2015; Brieant et al., 2021), while others have found delays (e.g., Hair et al., 2015; Keding et al., 2021), and still other others find differential associations (e.g., Beck et al., 2025; Sumner et al., 2019). Further research with longitudinal, repeated measures of pubertal development and brain structure is necessary to elucidate heterogeneity in the timing and tempo of developmental processes following adversity, as well as to explicate the consequences for key cognitive outcomes such as working memory.

Theoretical Perspectives

Alterations in the pace of development are thought to be responses to unmet developmental needs that optimize survival and reproductive fitness (Roubinov et al., 2021). Different theoretical perspectives have sought to explain the specific direction of these alterations (i.e., acceleration versus delay/deceleration) in pubertal maturation and brain development following adversity. Life History Theory is one perspective, grounded in evolutionary biology, which posits that slower or faster pace of development is shaped by early life experiences (Belsky, 2004). According to this theory, adversity exposure may prompt reprioritization of developmental strategies and more rapid maturation as an adaptation to

conditions of stress, thus optimizing survival and reproductive fitness. The stress acceleration hypothesis describes the putative neurobiological mechanisms that may be involved in accelerated maturation, particularly in emotion-related circuits in the brain (Callaghan & Tottenham, 2016). Much of the empirical support for this hypothesis has been derived from animal models (e.g., Callaghan & Richardson, 2014; Guadagno et al., 2020). However, emerging evidence suggests similar processes in humans. For example, accelerated pubertal development (Colich & McLaughlin, 2022; Thijssen et al., 2020) and earlier menarche in girls (Colich et al., 2020) has been observed following adversity and trauma, consistent with neurobiological evidence of more rapid maturation (Brieant et al., 2021; Thijssen et al., 2017).

On the other hand, adversity may instead contribute to delays or slowed progression in development. Chronic stress exposure can strain biological systems and has been associated with alterations such as loss of neurons, decreased dendritic branching, and decreased neurogenesis (Arnsten, 2009; Lupien et al., 2009), as well as delayed physical maturation (Mousikou et al., 2023). Some researchers suggest that these changes may represent adaptive responses aimed at conserving physiological resources in environments where essential needs, such as parental care and nutrition, are scarce (Roubinov et al., 2021). Taken together, these patterns of accelerated or delayed development likely represent key biological mechanisms mediating the relationship between adversity exposure and subsequent cognitive outcomes. However, despite growing interest in elucidating these pathways, empirical evidence has not provided consistent support for any one theory. It is possible that variability in findings reflects heterogeneity in children's responses to adversity, such that some children demonstrate accelerated patterns of development while others demonstrate delays. This idea is consistent with the concept of multifinality, where similar experiences lead to diverging developmental outcomes (Cicchetti & Rogosch, 1996). At

the same time, different developmental pathways may converge on similar outcomes, in line with the idea of equifinality. The goal of the present study is to elucidate average patterns of timing and tempo, while also acknowledging variability around those averages. In this way, we aim to lay the groundwork for identifying conditions under which acceleration or delay may emerge, and for whom.

Pubertal Development

Pubertal timing (i.e., relative maturation compared to same-aged, same-sexed peers) and tempo (i.e., pace of progression) varies across sex[1] and individuals. For example, girls on average start puberty 2-3-years before boys and often complete puberty faster than boys (Dorn & Beltz, 2023). At pubertal onset, youth experience increases in sex hormones such as estradiol, testosterone, and dehydroepiandrosterone (DHEA). These hormones cross the blood–brain barrier and act on brain regions with high concentrations of sex hormone receptors (e.g., the prefrontal cortex, hippocampus, and amygdala; Almey et al., 2015; Schulz et al., 2009). Because these brain regions are critical for executive functions such as working memory (Vijayakumar et al., 2018), emerging work has considered how pubertal timing and tempo may directly and indirectly shape cognitive development.

Pubertal timing has consistently been linked to adversity exposure and subsequent cognitive outcomes. For example, a recent meta-analysis suggested that adversity, in particular threat (i.e., trauma or violence exposure), was associated with earlier timing, especially in girls (Colich et al., 2020). Earlier timing has been associated with both accelerated and delayed cognitive development as well. Some studies report enhanced cognitive control and working

[1] Development is a biopsychosocial process; aspects of one's sex (male or female; assigned at birth) and gender (e.g., identity, sociocultural attitudes or behaviors) likely contribute to pubertal development, brain development, and experiences. For consistency, the authors use sex (boys: males, girls: females) throughout the present study, but recognize that this does not fully capture the diversity of experiences related to puberty and brain development.

memory skills in earlier developers (Chaku & Hoyt, 2019; Torvik et al., 2021) and others report worse cognitive outcomes in earlier developers, especially later in adolescence as their peers "catch-up" developmentally (Goering et al., 2023; Juraska & Willing, 2017). It is yet unclear whether earlier developers experience an initial cognitive boost that dissipates over time, whether effects are domain-general or domain-specific (see Luna et al., 2015; Del Giudice & Crespi, 2018) or whether slower, more protracted maturation ultimately supports more optimal cognitive development.

Pubertal tempo has rarely been considered in the extant literature given the limited number of studies with longitudinal data and differences in its definition and assessment across studies (see Cheng et al., 2020). Some evidence suggests that pubertal tempo is also associated with adversity exposure and cognitive outcomes, but the directionality of effects are not consistent and may be sex-specific. For example, exposure to abuse has been associated with slower tempo in girls (Hamlat et al., 2022), but lower SES in early childhood has been associated with faster tempo in boys (Chaku et al., 2024). Other studies have found no links between adversity exposure and tempo (e.g., Ho et al., 2024; Suglia et al., 2020), or differential associations (e.g., MacSweeney et al., 2025). Even fewer studies have linked pubertal tempo to cognitive outcomes, though there is some evidence that faster tempo is associated with worse working memory and decision-making especially in lower SES contexts (Chaku et al., 2024). It remains unclear whether differences in the literature are due to the operationalization of tempo, the age range of the sample, the measure of adversity, or actual differences between sexes.

Neurodevelopment

Typical gray matter development follows a protracted, non-linear trajectory (Gennatas et al., 2017; Mills et al., 2016; Tamnes et al., 2017), increasing across childhood and then

decreasing across adolescence, reflecting multiple cellular-level processes such as pruning, myelination, and synaptic remodeling. Gray matter volume (GMV) provides a metric to capture global morphometry changes during a critical developmental window; more negative GMV change reflects relatively more rapid macrostructural maturation over time, whereas less negative change indicates relatively slower or delayed maturation. Recent evidence from the Adolescent Brain Cognitive Development (ABCD) Study indicates that GMV is a stronger predictor of cognitive performance relative to other structural metrics such as cortical thickness and surface area (Michel et al., 2024). Importantly, there is considerable individual-level variability in total GMV trajectories during adolescence (Mills et al., 2021; Shi et al., 2024). This variability may be attributed to a variety of factors, including genetic variation as well as differences in environmental influences (Brouwer et al., 2017; Dow-Edwards et al., 2019).

Adverse experiences tend to be associated with lower GMV (Paquola et al., 2016; Walsh et al., 2014) and emerging longitudinal evidence also suggests associations with rate of change in cortical and subcortical GMV (Barch et al., 2022; Breslin et al., 2024). However, the directionality of findings has been mixed across studies (Hakamata et al., 2022). Understanding these patterns is especially important because GMV maturation supports key cognitive abilities across development. Specifically, prior studies have shown a negative association between different metrics of GMV and general cognitive performance (e.g., de Chastelaine et al., 2023; Smolker et al., 2015). In the present study, we include working memory as a well-validated indicator of cognitive functioning in order to assess whether individual differences in timing and tempo are meaningfully associated with real-world outcomes.

The present study

Pubertal onset and its progression coincides with, and is associated with, changes in brain development across adolescence. For example, earlier developers have higher brain age and the rate of change in pubertal development is associated with the rate of change in brain age (Holm et al., 2023). Additionally, studies have found a correlation between pubertal status and cortical surface area changes in boys (Beck et al., 2023). Thus, adversity-related shifts in developmental timing – whether accelerating or delaying maturation – may be a key environmental mechanism that simultaneously shapes both pubertal and brain development. Clarifying individual differences in timing and tempo of pubertal maturation and brain development is crucial as these developmental processes shape cognitive abilities during childhood and adolescence (Blakemore & Choudhury, 2006; Tamnes et al., 2013).

Critically, many prior studies have been limited by the lack of longitudinal data capturing pubertal development and neurodevelopment in youth over time. The ABCD Study offers a unique opportunity to test these questions in a large sample. Currently, the study includes at least three time points of both pubertal and neuroimaging data, the minimum requirement for identifying a linear slope and estimating within-person change over time (Parsons & McCormick, 2024). The present study will assess whether (1) associations between adversity and working memory are mediated by pubertal timing and brain maturation, and (2) whether these associations differ by sex. A more granular description of the specific questions and hypotheses, as well as the type of analyses, effect of interest, and threshold for evidence, are elaborated on in Table 1. Given the limited and inconsistent extant literature, we do not propose directional hypotheses related to pubertal tempo and pace of brain development, and these parts of the analyses will be considered exploratory. We take a global approach to examine broad associations among cumulative adversity, pubertal status, and brain volume. While these

cumulative indices simplify complex experiences, establishing robust general patterns is a first step before parsing heterogeneity across domains of adversity and their distinct neurodevelopmental impacts.

Table 1. List of research questions and corresponding hypotheses.

Research Question (RQ)	Prediction	Type of analysis	Effect of interest	Threshold for evidence	Supported?
RQ1. What are the associations between pubertal development and brain maturation?	H1: Pubertal development will be associated with brain maturation. Specifically, earlier pubertal timing (age at midpoint of puberty) will be associated with: (a) more advanced brain maturation (i.e., lower GMV) at baseline (intercept), and (b) faster brain maturation (slope, i.e., steeper declines in GMV over time)	Path analysis	Significant regression coefficients (adjusted $p \leq .05$)	$\beta \geq .01$	
RQ2: Does adversity predict pubertal development and brain maturation?	H2: Greater adversity exposure will be associated with: (a) earlier pubertal timing and; (b) more advanced brain maturation at baseline	Path analysis	Significant regression coefficients (adjusted $p \leq .05$)	$\beta \geq .01$	
RQ3: Does adversity, pubertal development, and brain maturation predict working memory?	H3: Adversity, pubertal development, and brain maturation will be associated with working memory. Specifically: (a) more adversity will be associated with better working memory (b) earlier pubertal timing will be associated with better working memory (c) more advanced brain maturation will be associated with better working memory	Path analysis	Significant regression coefficients (adjusted $p \leq .05$)	$\beta \geq .01$	
RQ4: Do pubertal development and brain maturation mediate associations between adversity and working memory?	H4: Associations between adversity and working memory will be mediated by pubertal development and brain maturation. Specifically: (a) earlier pubertal timing will mediate associations between adversity and working memory (path a_1*b_1) (b) more advanced brain maturation will mediate associations between adversity and working memory (path a_2*b_2)	Parallel mediation analysis	Significant indirect effects: adversity to the putative mediator (path 'a'), putative mediator to working memory (path 'b'), calculated using the product-of-coefficients method (a*b)	Bias-corrected bootstrapped confidence intervals that do not include zero	
RQ5: Do associations between adversity, pubertal	H5: Effects will be more pronounced in girls than boys.	Wald tests	A significant wald test (adjusted $p \leq .05$) between	$\beta \geq .01$	

development, brain maturation, and working memory differ by sex?	coefficients for boys versus girls

Method

Data will be drawn from the ABCD study, a longitudinal study of 11,876 youth (~50% male; ~50% female) from 21 research sites across the United States, beginning when youth were 9-10-years old. We will use data release 6.0 which includes data through the six-year follow-up (including seven timepoints of puberty data and four neuroimaging timepoints). Information on recruitment, enrollment, and study retention are outlined elsewhere (Casey et al., 2018; Garavan et al., 2018) and detailed information on the ABCD study design and experimental procedures can be found at https://abcdstudy.org/. Institutional Review Board (IRB) approval was obtained for each site before data collection; all parents provided written informed consent and all youth provided assent.

Participants

The analytic sample will include all participants with available data. The 6.0 data release includes approximately half of the six-year follow-up data (n = 5,056). Exclusion criteria for the study included lack of English language proficiency and any severe sensory intellectual, medical, or neurological issues (see all exclusion criteria in ABCD release notes; Saragosa-Harris et al., 2022).

Measures

We will use data from baseline (9-10-years old) to derive adversity exposure, data from baseline (9-10-years-old) through the six-year follow-up (15-16-years-old) to derive pubertal timing and tempo, and data from baseline, the second-year follow-up, fourth-year follow up, and six-year follow-up to derive trajectories of neurodevelopment. Working memory outcomes will be drawn from the six-year follow-up.

Adversity exposure. Overall adversity exposure will be indexed by a composite score of items (ABCD-ELA^{+}) from baseline measures that capture a range of adverse experiences, following methods from Breslin et al. (2025). In this approach, items are derived from nine different measures in the ABCD Study, including the Kiddie Schedule for Affective Disorders and Schizophrenia (Kaufman et al., 1997), the PhenX neighborhood crime and safety scale (Mujahid et al., 2007), and the Children's Report of Parent Behavior Inventory (CRPBI; Schaefer, 1965) and include a mix of caregiver-report and child self-report. The items are sorted into 14 adversity domains such as physical abuse, parental incarceration, violence, emotional neglect, and access to resources. A score of 1 is assigned if either the caregiver or the youth participant endorsed any of the variables in that domain, and then a sum score across all domains is computed. Higher composite scores will reflect a higher degree of adversity exposure. Further details on the computation of this variable and the R code that will be used can be found at: https://github.com/karalk07/abcd-ela.

Pubertal timing and tempo. Pubertal development will be assessed via the Pubertal Development Scale (PDS; Petersen et al., 1988), a well-known and validated measure of pubertal status administered yearly to both children and their parents. The PDS is composed of five items (three sex-neutral items and two sex-specific items) assessing common pubertal milestones (e.g., breast development for girls, voice changes for boys) rated on a four-point scale where 1 = *has not yet started* and 4 = *seems complete*. Items will be averaged at each time point to create a composite score (ranging from 1 - 4). The composite scores will be used to derive individual estimates of timing and tempo (described below). There is substantial debate (see Dorn & Beltz, 2023) regarding whether to use the child-reported or parent-reported (or the average of) PDS items. In the current study, the parent report will be used in the focal analyses because some

evidence suggests that adolescents tend to overestimate their development at the start of puberty (Shirtcliff et al., 2009). Analyses will be re-run using the child-reported PDS as a sensitivity analysis.

Neurodevelopment. Structural brain maturation will be indexed by cortical and subcortical gray matter volume (GMV) at each available time point. All data were pre-processed and analyzed by the consortium's data analytic core (Hagler et al., 2019). Imaging data that does not meet quality control standards (i.e., recommended for exclusion by the analytic core) will be coded as missing. Scanning protocols were harmonized across each of the 21 study sites and GMV was estimated for each participant at each time point in FreeSurfer v6.0. Cortical and subcortical GMV will be tested in separate models to examine similarities or differences across these metrics. Complete details of data acquisition procedures and preprocessing methods are described elsewhere (Casey et al., 2018; Hagler et al., 2019).

Working memory. Working memory at baseline and the six-year follow-up will be indexed by performance on the List Sorting Task from the National Institutes of Health Cognitive Toolbox (NIH-TB; Weintraub et al., 2013). The NIH toolbox consists of several tasks that evaluate memory, executive function, language, and other higher-order cognitive processes (Taylor et al., 2022). The List Sorting Working Memory Test requires participants to sequence task stimuli based on category membership and perceptual characteristics. Specifically, participants were presented with computerized, illustrated pictures depicting an item and its auditory name for two seconds. Participants were then asked to remember the stimuli and then verbally repeat them from smallest to largest. The repeated series increased in length and the longest set size will be used as an indicator of working memory (Rosenberg et al., 2020)

Analytic Plan

Data analyses will take place over three phases. First, descriptive statistics (i.e., means, standard deviations, and frequency distributions), normality, and sample attrition will be assessed; then, pubertal development and brain maturation indices will be estimated separately using mixed effects models and latent growth modeling respectively (see Figure 1a for a conceptual model). Second, path analyses will be used to describe associations between adversity, pubertal development, brain maturation, and working memory (see Figure 1b for a conceptual model). Finally, parallel mediation analysis will be used to understand whether pubertal development and brain maturation mediate associations between adversity and working memory, and Wald tests will be used to investigate whether associations differ by sex (see Figure 1c for a conceptual model). Each phase is described in the following sections.

Data Cleaning and Variable Preparation

Puberty data will first be examined to ensure that only reliable and valid data will be included. Following previous research (Chaku et al., 2024), only adolescents with at least two assessments will be included in subsequent analyses. For neuroimaging data, all scans that meet quality control standards will be included. Longitudinal imaging data will be harmonized using longComBat to address site and scanner effects and reduce variability due to technical artifacts (Beer et al., 2020)

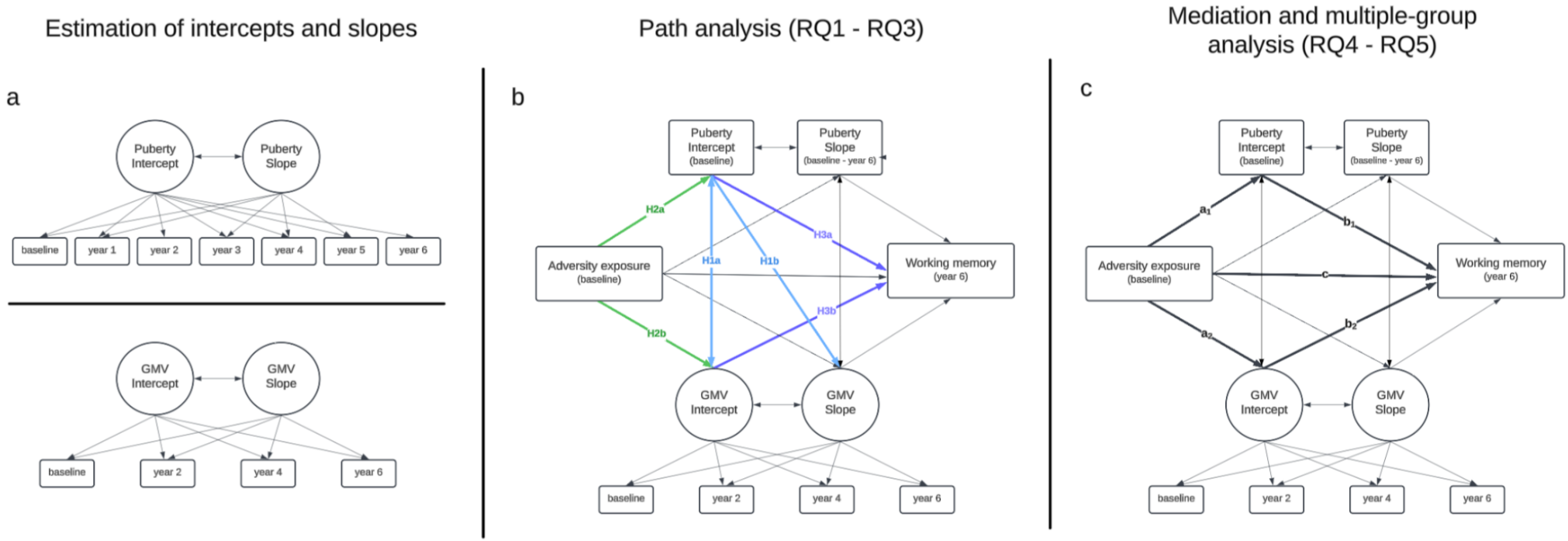

Figure 1. Conceptual models depicting the phases of analysis and hypothesized associations among adversity exposure, brain and pubertal development, and cognitive outcomes. a) Estimation of intercept and slope for puberty (mixed-effect model) and gray matter volume (latent growth curve model). b) Path analysis integrating adversity, puberty intercept and slope (extracted from mixed-effect models and entered as manifest variables), GMV intercept and slope (latent variables), and working memory outcomes. c) Hypothesized indirect effect pathways. For each phase, two separate models will be conducted for cortical vs subcortical GMV.

Note. Circles represent latent variables, rectangles represent manifest variables. GMV = gray matter volume. H1a-H3b = hypothesized associations, detailed in Table 1. a1 - c = mediation paths.

Unconditional models of puberty and brain maturation

Pubertal timing and tempo. Estimates of pubertal timing and tempo will be derived in SAS (SAS Institute, Cary NC) using nonlinear mixed-effects models, separately for boys and girls. First, a logistic growth curve function will be fit to the data. In a logistic growth curve model, timing is conceptualized as age at the midpoint of puberty, or PDS = 2.5, with higher scores indicating later timing or older age at PDS = 2.5. Tempo is conceptualized as the rate of development at the midpoint of puberty with higher scores indicating faster tempo, or progressing through more pubertal stages per year at the midpoint of puberty. Models will be run separately for boys and girls to obtain fixed effects (or sample-level estimates of timing and tempo). Then, Bayes empirical estimates will be used to extract individual-level estimates. Once Bayes empirical estimates are extracted, timing will be represented by age at the midpoint of puberty and tempo will be represented by the number of pubertal stages completed per year. With seven timepoints, the logistic model is likely to converge, but if it does not, a linear model will be specified to obtain estimates of timing and tempo, and individual-level estimates will be extracted.

Neurodevelopment. In Mplus version 8.0 (Muthén & Muthén, 2017), we will use latent growth curve modeling to estimate the intercept (initial level) and slope (rate of change) of 1) cortical and 2) subcortical GMV. By examining the mean and variance of the slope factor, we will be able to delineate increases or decreases in GMV across adolescence as well as individual differences in these changes. We will first fit a series of unconditional growth models to estimate the shape of the data: 1) no growth (reflecting stability over time), 2) linear growth (reflecting linear increases or decreases over time), 3) quadratic growth (reflective non-linear changes over time), and 4) latent-basis growth (allowing the data to define the shape of growth without

specifying a particular functional form). The most parsimonious model with the best fit to the data (determined using chi-square difference tests for model fit comparisons) will be selected as the final model. For all structural equation models, RMSEA values of less than .05 will be considered a close fit while values less than .08 will be considered a reasonable fit (Browne & Cudeck, 1992). CFI values of greater than .90 will be considered an acceptable fit while values greater than .95 will be considered an excellent fit (Bentler, 1990). We will estimate these growth patterns separately for boys and girls to account for potential sex differences.

Path analyses

To test the first three research questions, we will estimate a series of path analyses in MPlus to examine associations between adversity, pubertal development (timing and tempo), brain maturation (GMV at intercept and change in GMV over time), and cognition (working memory; see Figure 1b) separately by sex. First, paths will be estimated between pubertal timing and brain maturation at baseline (H1a) and between pubertal timing and change in brain maturation over time (H1b) as well as between all other possible paths (e.g., between pubertal timing and tempo and between pubertal tempo and change in brain maturation) to investigate additional exploratory effects. Then, paths will be estimated between adversity and pubertal timing (H2a) and between adversity and brain maturation at baseline (H2b) as well as between all other possible paths (e.g., between adversity and pubertal tempo and between adversity and change in brain maturation) to investigate additional exploratory effects. Finally, paths will be estimated between adversity and working memory (H3a), between pubertal timing and working memory (H3b) and between brain maturation at baseline and working memory (H3c) as well as between all other possible paths (e.g., between pubertal tempo and working memory and

between change in brain maturation and working memory) to investigate additional exploratory effects.

Adversity, pubertal timing and tempo, and working memory will be manifest variables and GMV timing and tempo will be latent variables. To account for the complex sampling structure in the ABCD Study, we will specify stratification by study site and clustering of siblings within families. Missing data will be estimated using Full Information Maximum Likelihood (FIML) with robust standard errors (MLR). To control for multiple comparisons, *p*-values from planned tests will be adjusted using the Benjamini–Hochberg False Discovery Rate (FDR) procedure (Benjamini & Hochberg, 1995); the FDR procedure will be applied separately within each family of related tests (RQ1-RQ3: path analyses, RQ4: parallel mediation analysis, RQ5: Wald tests) separately for girls and boys using a false discovery rate of 0.05. Standardized coefficients equal to or exceeding $\beta = .01$ will be interpreted as a meaningful effect size following previous research using ABCD data (Saragosa-Harris, 2017).

Parallel mediation analyses

To test the final two research questions, we will use parallel mediation analyses in MPlus to assess whether pubertal development and brain maturation mediated the association between adversity and working memory in a single model as recommended in previous research (see Fritz & MacKinnon, 2008). Indirect effects will be computed using the product-of-coefficients approach (i.e., $a \times b$), where path *a* represents the association between adversity and each putative mediator (pubertal timing: path a_1, brain maturation at baseline: path a_2) and path *b* represents the association between each putative mediator and working memory (pubertal timing: path b_1, brain maturation at baseline: path b_2), controlling for adversity (see Figure 1c). Additional indirect effects may be specified but will only be considered as exploratory. We will

formally test whether associations differ between boys and girls using a Wald test to compare unstandardized regression coefficients. Specifically, we will compute a z-statistic for the difference between beta estimates from the boys model vs the girls model. Significant *p* values will indicate that the strength of association differs by sex. Confidence intervals for total, direct, and indirect effects will be derived using bias-corrected nonparametric percentile bootstrapping with 10,000 resamples (see Fritz & MacKinnon, 2008; Hayes, 2015). Indirect effects will be considered statistically significant if the 95% bootstrapped confidence interval for the indirect effect does not include zero.

Sensitivity analyses

The primary analyses do not include baseline working memory as a covariate because doing so may produce biased estimates (i.e., regression to the mean, conditioning on a collider; see McCormick & Bauer, 2024); however, in exploratory analyses, we will estimate a latent change score model to assess the effect of adversity, pubertal development, and brain development on latent change in working memory (Castro-Schilo and Grimm, 2018). This approach mitigates some of the limitations of residualized and difference score models, allowing us to test aspects of directionality. To assess potential regression to the mean, we will additionally reverse the direction of the latent change factor (from later to earlier measurement occasions) to determine whether the sign and magnitude of effects remain stable (see McCormick & Bauer, 2024).

Summary and Timeline

Leading theoretical perspectives suggest that adversity exposure affects not only the degree of pubertal maturation and brain volume at a single time point, but also the developmental trajectories of these processes over time. However, empirical tests of these theoretical models

have been limited due to a lack of longitudinal, multimodal data, and the analytic and methodological complexity involved. The proposed analyses will capitalize on the unprecedented longitudinal data from the ABCD Study to clarify the effects of adversity on pubertal and GMV development, as well as subsequent impacts on working memory. By using a registered report format and providing open code and data, this work will ensure replicability and reproducibility, offering a critical contribution to a literature marked by heterogeneous analytic approaches and inconsistent findings. Findings will also provide an essential foundation for future investigations that seek to disentangle the effects of different types of adversity on timing and tempo, as well as possible region-specific impacts. The authors have published prior studies using ABCD Study data and are familiar with some of the variables in the analysis. The authors have not constructed or analyzed any of the variables in the ways described in the present manuscript. Data cleaning and analyses will be completed by the end of November 2025, and the Stage 2 manuscript will be submitted by the end of February 2026.

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